

Generalized Self-Similar Solutions of Subsonic Marshak Waves

Theory and Numerical Validation Structure

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Inertial Confinement Fusion (ICF)

- Hohlraum Modeling Crucial for radiation temperature prediction and experimental measurement accuracy.
- Energy Delivery Key to analyzing laser–walls coupling and refining pulse shaping techniques.
- Historical Context Builds upon classical foundations established by *Lindl (1995)*, *Kauffman (1995)*, and *Rosen (2005)*.

Adding Shay & Menahem's suggestions

- Slides about the Aluminum article...

The Theory Of Subsonic Heat Waves

Physics Of The Problem

- We treat a semi-infinite wall subjected to high-temperature boundary heating.
- Dynamic equilibrium is mediated strictly via **photon transport**.
- In the **optically-thick limit**, an exceptionally sharp temperature front develops.

The Transfer Equation Coupled to Euler Equations

- Coupled Boltzmann & internal energy equation.
- The diffusion equation under the gray approximation.
- Euler equations for the hydrodynamic evolution of the material - in two words.
- The EOS for different materials fitted by H & R (2003 or previously).